Laxmi Charitable Trust's

Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce

Department of Information Technology (B.Sc.I.T Semester V)

Data Visualization with Power BI and Tableau

Practical – IV

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Class: TYIT	Batch: 01
Date of Assignment: 20/08/2026	Date/Time of Submission: 20/08/2026

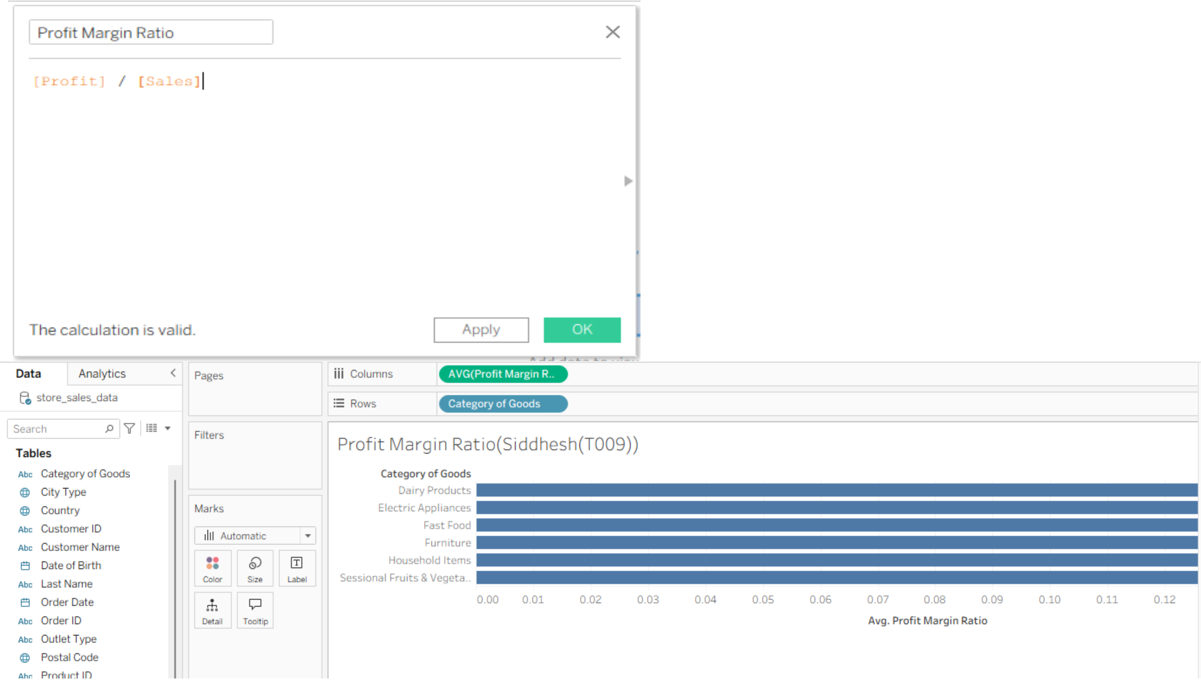
Performing Data Analysis Using Calculations in Tableau

a) Create Basic Calculated Fields Using Formulas

Steps:

1. Open store_sales_data.csv in Tableau.
2. Go to **Data Pane → Create Calculated Field**.
3. Name it **Profit Margin Ratio**.
4. Enter:
   [Profit] / [Sales]
5. Click **OK**.
6. Add Category of Goods → Rows and Profit Margin Ratio → Columns.
7. Change the calculation to **Average**.

Output:



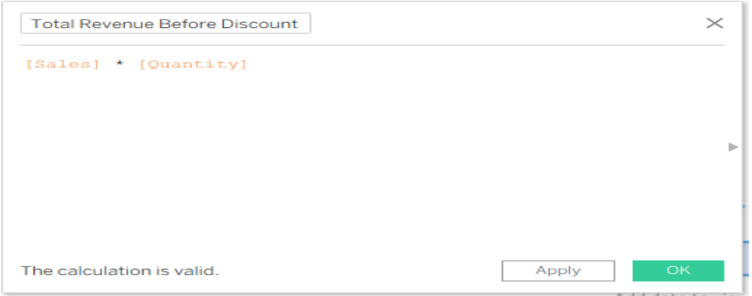
b) Perform Row-Level Calculations

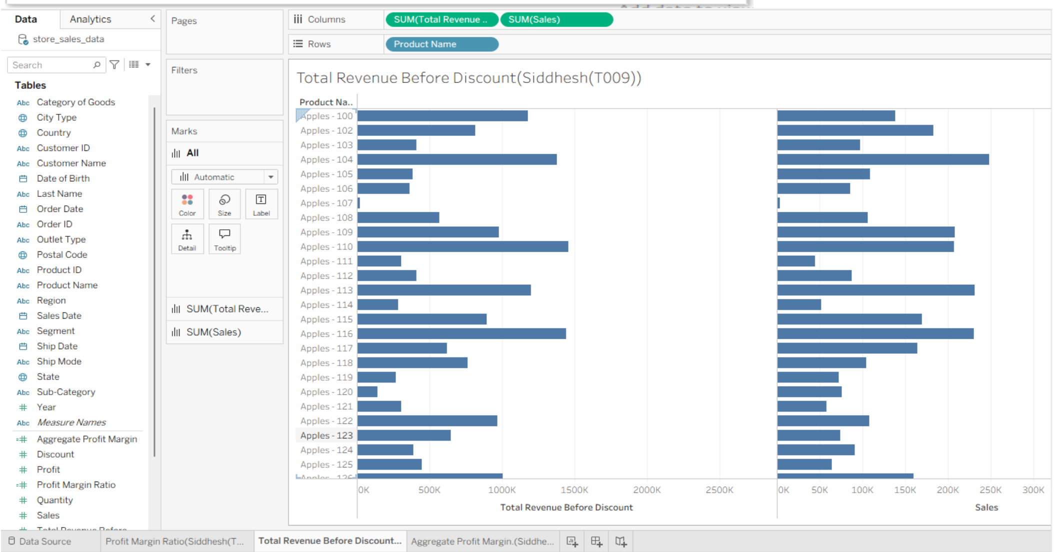
Steps:

1. Create a new **Calculated Field**.
2. Name it **Total Revenue Before Discount**.
3. Enter:
   [Sales] * [Quantity]
4. Click **OK**.

5. Add Product Name → Rows.
6. Add Total Revenue Before Discount → Columns.
7. Add Sales → Columns for comparison.

Output:



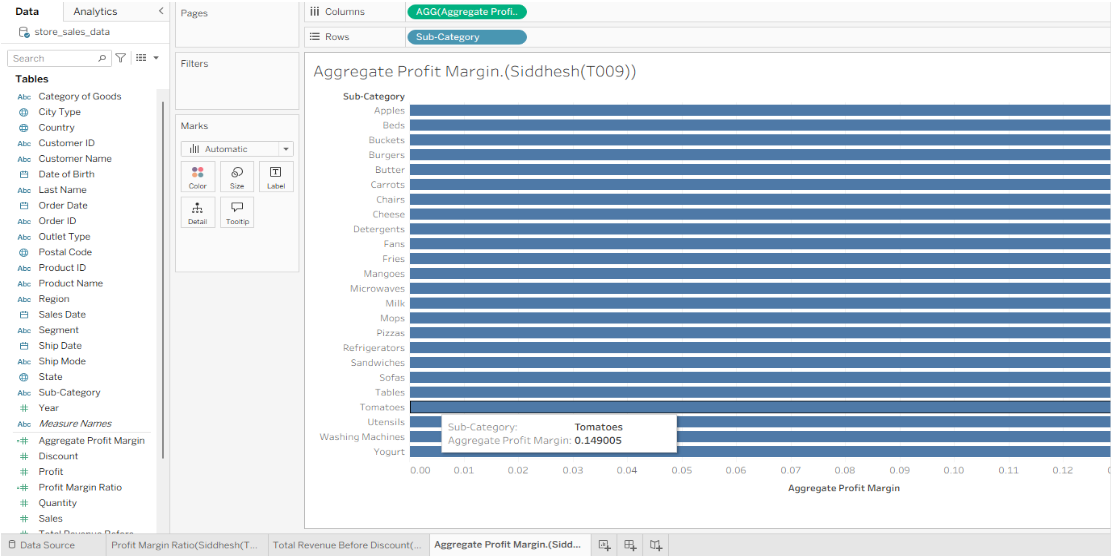


c) Perform Aggregate-Level Calculations

Steps:

1. Create a new **Calculated Field**.
2. Name it **Aggregate Profit Margin**.
3. Enter:
   SUM([Profit]) / SUM([Sales])
4. Click **OK**.
5. Add Sub-Category → Rows.
6. Add Aggregate Profit Margin → Columns.

Output:

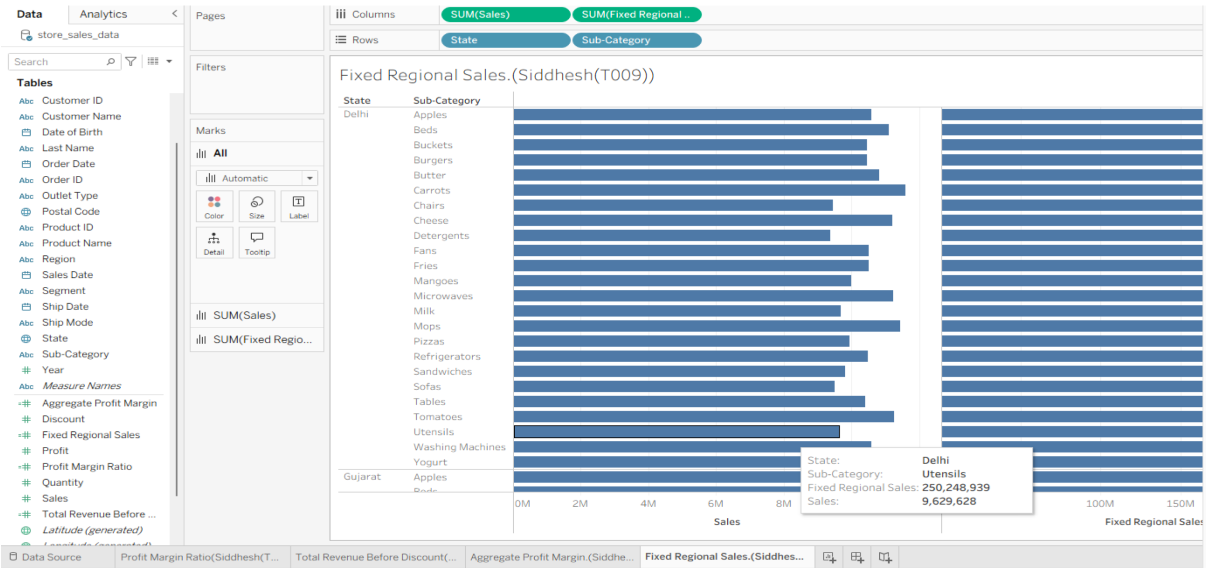


d) Implement FIXED Level of Detail (LOD) Calculations

Steps:

1. Create a new **Calculated Field**.
2. Name it **Fixed Regional Sales**.
3. Enter:
   { FIXED [State] : SUM([Sales]) }
4. Click **OK**.
5. Add State → Rows.
6. Add Sub-Category → Rows.
7. Add Sales and Fixed Regional Sales → Columns.

Output:



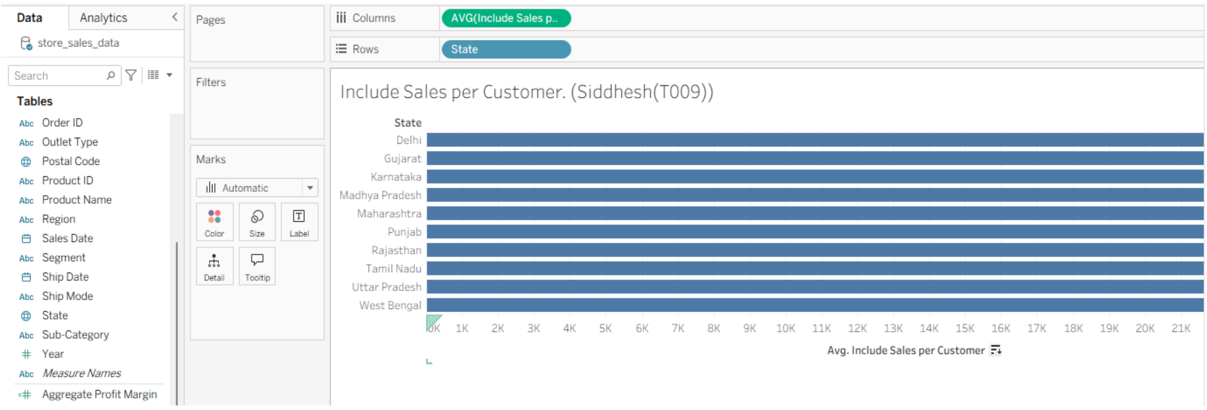
e) Use INCLUDE and EXCLUDE LOD Expressions

Steps:

INCLUDE LOD:

1. Create a calculated field named **Include Sales per Customer**.
2. Enter:
   { INCLUDE [Customer ID] : SUM([Sales]) }
3. Add State → Rows.
4. Add Include Sales per Customer → Columns.
5. Change the aggregation to **Average**.

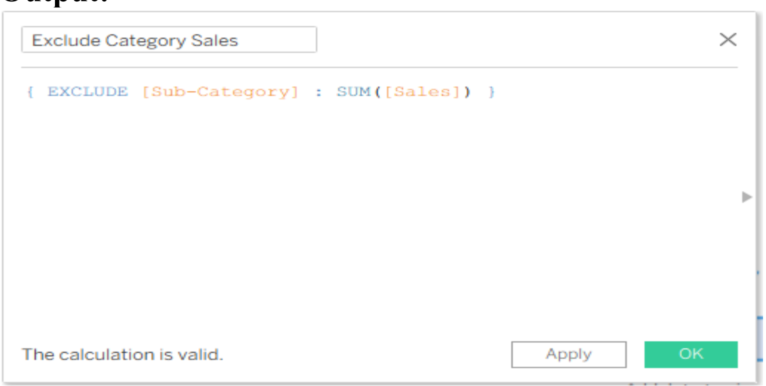
Output:

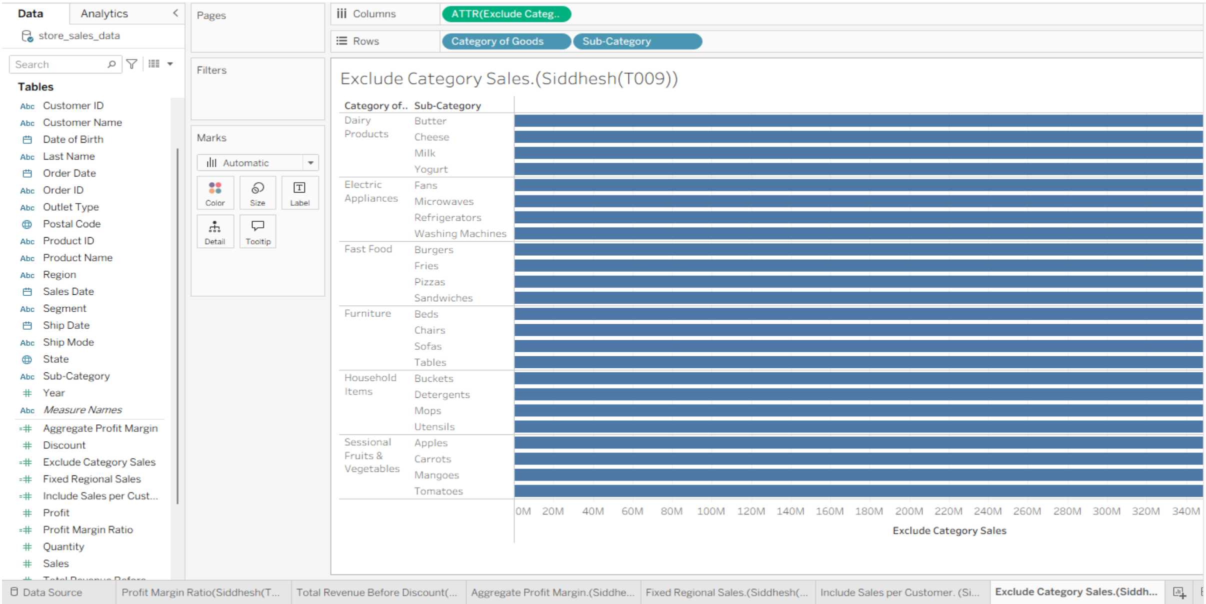


EXCLUDE LOD:

1. Create a calculated field named **Exclude Category Sales**.
2. Enter:
   { EXCLUDE [Sub-Category] : SUM([Sales]) }
3. Add Category of Goods and Sub-Category → Rows.
4. Add Exclude Category Sales → Columns.

Output:



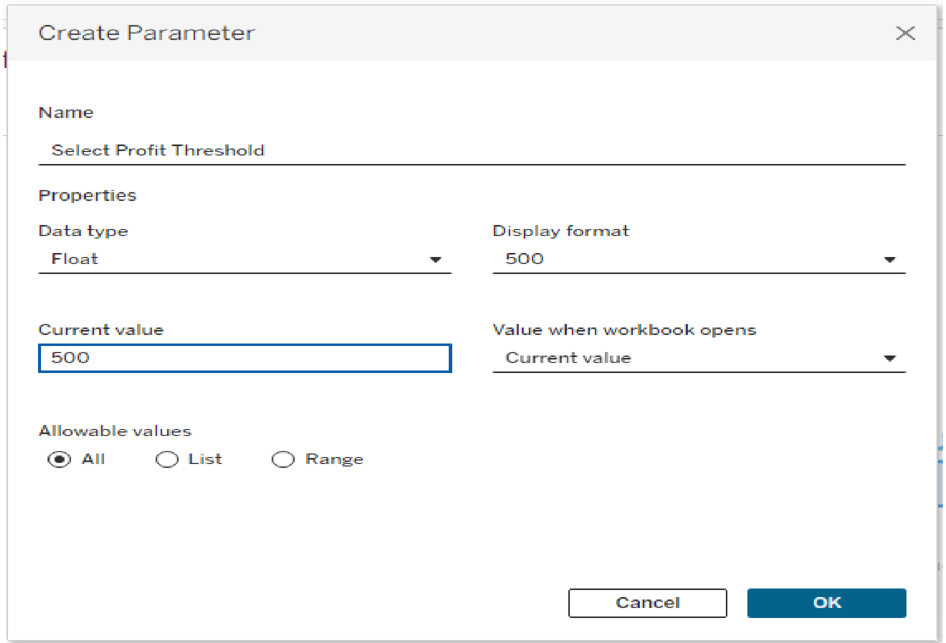


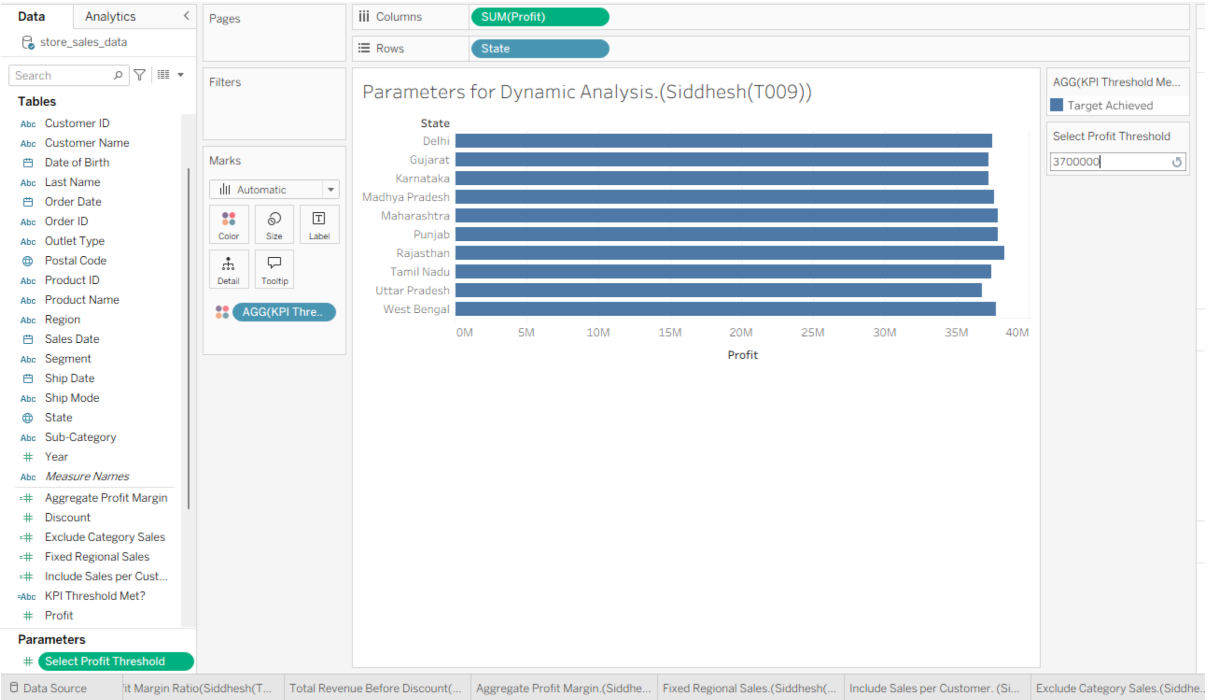
f) Create and Use Parameters for Dynamic Analysis

Steps:

1. Go to **Data Pane → Create Parameter**.
2. Name it **Select Profit Threshold**.
3. Set Data Type to **Float/Integer**.
4. Set Current Value to **500**.
5. Create a calculated field named **KPI Threshold Met?**
6. Enter:
   IF SUM([Profit]) >= [Select Profit Threshold] THEN "Target Achieved" ELSE "Below Target" END
7. Add State → Rows and Profit → Columns.
8. Add KPI Threshold Met? → Color.
9. Select **Show Parameter Control**.

Output:





g) Apply Calculations to Fix Data Issues and Enhance Insights

Steps:

Handling Null Values:

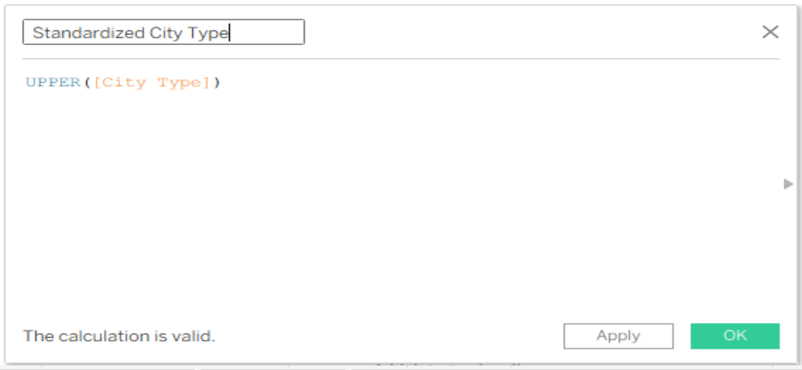
1. Create a calculated field named **Cleaned Discount**.
2. Enter:
   IFNULL([Discount], 0)
3. Click **OK**.

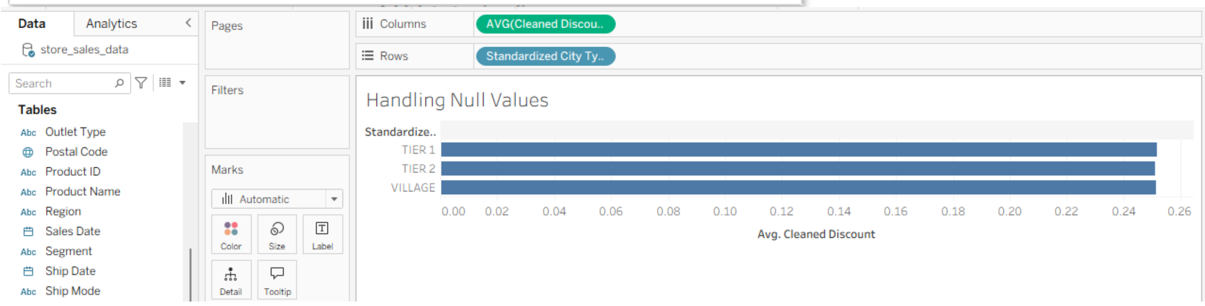
Standardizing Text:

1. Create a calculated field named **Standardized City Type**.
2. Enter:
   UPPER([City Type])
3. Click **OK**.
4. Add Standardized City Type → Rows.
5. Add Cleaned Discount → Columns.
6. Change aggregation to **Average**.

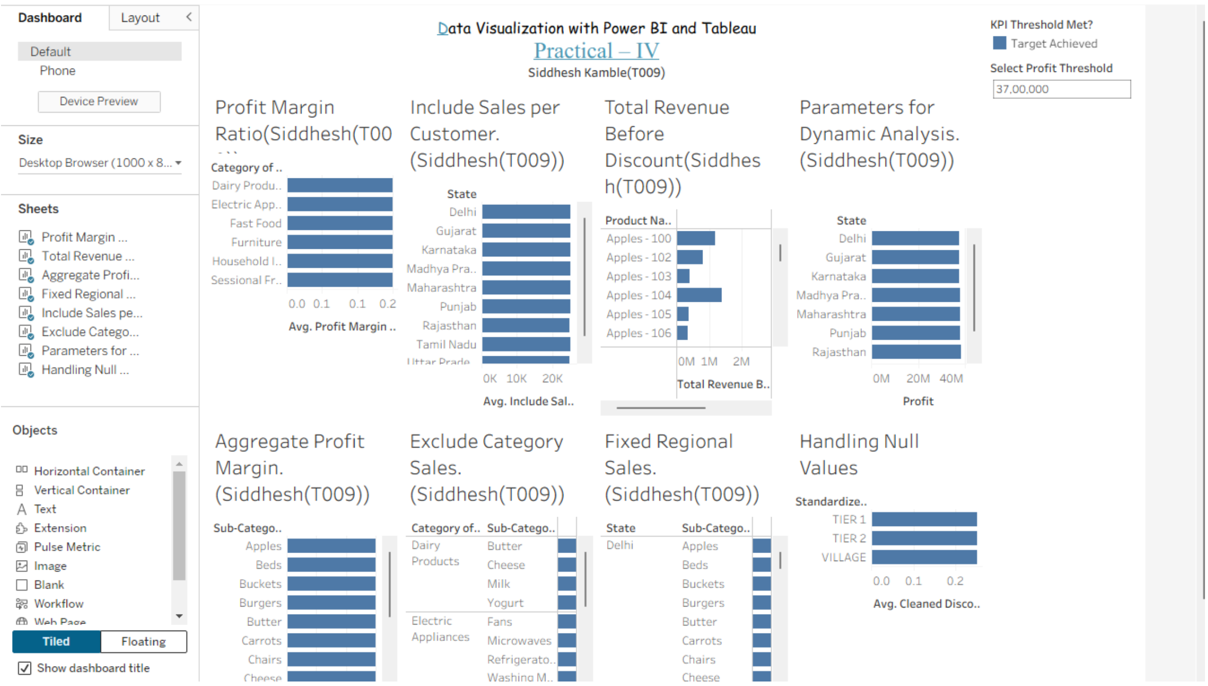
Output:


Cleaned Discount
IFNULL([Discount], 0)
The calculation is valid.
Apply
OK


Standardized City Type
UPPER([City Type])
The calculation is valid.
Apply
OK


Data
Analytics
store_sales_data
Tables
Outlet Type
Postal Code
Product ID
Product Name
Region
Sales Date
Segment
Ship Date
Ship Mode
Pages
Filters
Marks
Automatic
Columns
AVG(Cleaned Discou..
Rows
Standardized City Ty..
Handling Null Values
Standardize..
TIER 1
TIER 2
VILLAGE
Avg. Cleaned Discount

Dashboard :


Dashboard
Layout
Default
Phone
Device Preview
Size
Desktop Browser (1000 x 8...
Sheets
Profit Margin ...
Total Revenue ...
Aggregate Profi...
Fixed Regional ...
Include Sales pe...
Exclude Catego...
Parameters for ...
Handling Null ...
Objects
Horizontal Container
Vertical Container
Text
Extension
Pulse Metric
Image
Blank
Workflow
Tiled
Floating
Show dashboard title
Data Visualization with Power BI and Tableau
Practical – IV
Siddhesh Kamble(T009)
KPI Threshold Met?
Target Achieved
Select Profit Threshold
37,00,000
Profit Margin Ratio(Siddhesh(T00
Include Sales per Customer. (Siddhesh(T009))
Total Revenue Before Discount(Siddhesh(T009))
Parameters for Dynamic Analysis. (Siddhesh(T009))
Aggregate Profit Margin. (Siddhesh(T009))
Exclude Category Sales. (Siddhesh(T009))
Fixed Regional Sales. (Siddhesh(T009))
Handling Null Values